

Fund Strategies and Performance Evaluation

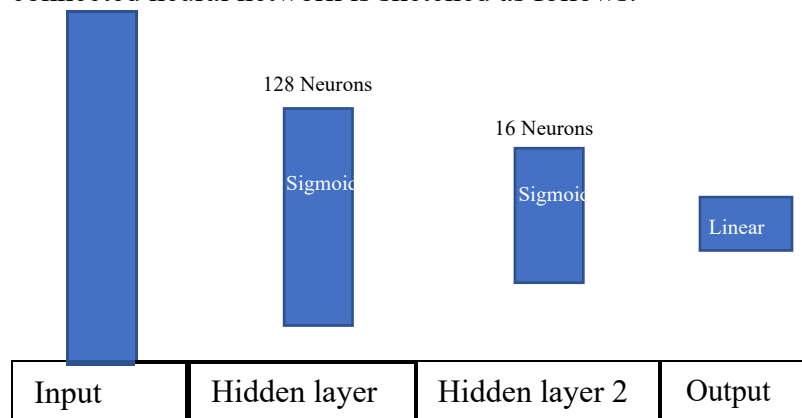
Assignment (HW6) Extra credit 10%

Objectives:

In this assignment, you are asked to create a fully connected neuron network to predict monthly return using the signals provided in “rpsdata_rfs_cleaned_2000.csv”. You need to create a dynamic fully connected neuron network model using relatively recent historical data. Then you will use the predicted returns of the dynamic model to construct an equally weighted long- short portfolio for each month.

Detailed instructions:

- 1) Follow the procedure in “Demo Random Stock Selection.ipynb” to select 100 stocks with the signals. Use random seed 2021 when sampling stocks.
- 2) Create a dynamic fully connected neuron network to predict monthly return using the signals. Select the lengths of training and testing periods to be 120 and 12. When each testing period ends, the model must be updated using all historical data. The structure of the fully connected neural network is sketched as follows.



When you fit the model, use the following setting:

All random seeds=2021

Threshold for removing columns with too many NaNs: 20%

Epochs=50

Optimizer: Adam with learning rate 0.01

Verbose=0

Batch_size=1000

validation_split: Do not include this parameter in model.fit (we can't afford the long runtime for doing validation in each training period.)

- 3) Use the predicted returns of the dynamic model to construct an equally weighted long-short portfolio for each month. Select the top 10 stocks and the bottom 10 stocks to form equal-weighted long and short portfolios, respectively. Report the mean and the standard deviation of the portfolio returns from all testing months. Report the cumulative return up to Dec 2019.